Calibration of the Riverside Weir Flow Gauges

Prepared by the Hydrometry Section, June 1996

Abstract

The three ultrasonic level gauges on the Riverside weir were checked against current-meter gaugings between March and May 1996. Two gauges read within 4 mm of the reference staff; the third drifted by 23 mm after the April flood and was re-zeroed. Discharge computed from the rated weir equation agrees with the gaugings to within 3.1 percent across the observed range of 1.2 to 46 cubic meters per second. We recommend quarterly staff checks and a new gauging after every flow above 40 cubic meters per second.

1 Background

The Riverside weir is a broad-crested concrete weir 18.0 m wide, built in 1958 to hold the town reach at a steady level. Since 1987 it has also served as the authority's main flow station. Its rating converts the upstream head over the crest into discharge with the standard broad-crested weir equation given below, whose coefficient was fitted to gaugings made in 1988.

The level gauges were replaced in 1994. Each measures the time of flight of an ultrasonic pulse to the water surface and back, corrects it for air temperature, and logs the level every fifteen minutes. Their manufacturer quotes an accuracy of 2 mm, but the mountings on the old footbridge move with temperature, and debris in the April flood struck the bridge.

$$Q = C \cdot b \cdot h^{3/2}, \quad h \ge 0.05\ \text{m} \qquad (1)$$

where Q is the discharge, b the crest width, h the head and C the coefficient (1.62 here).

2 Method

On eight visits the field team read the staff gauge, measured the discharge with a current meter at twenty verticals, and noted the three logged levels for the same quarter hour. The gaugings span the range of flows usual in spring.

Table 1 Gauge readings and discharges, 1996 (Q in cubic meters per second)

Date	Staff(m)	G1(m)	G2(m)	G3(m)	Q gauged	Q rated
03/04/96	0.212	0.214	0.211	0.213	1.24	1.21
03/18/96	0.488	0.490	0.486	0.489	4.31	4.40
04/02/96	0.951	0.953	0.949	0.950	11.72	11.93
04/09/96	1.734	1.737	1.731	1.739	28.65	29.40
04/16/96	2.312	2.315	2.309	2.335	45.90	46.28
04/30/96	1.106	1.108	1.104	1.129	14.80	14.96
05/14/96	0.637	0.639	0.635	0.614	6.48	6.57
05/28/96	0.402	0.404	0.400	0.403	3.25	3.29

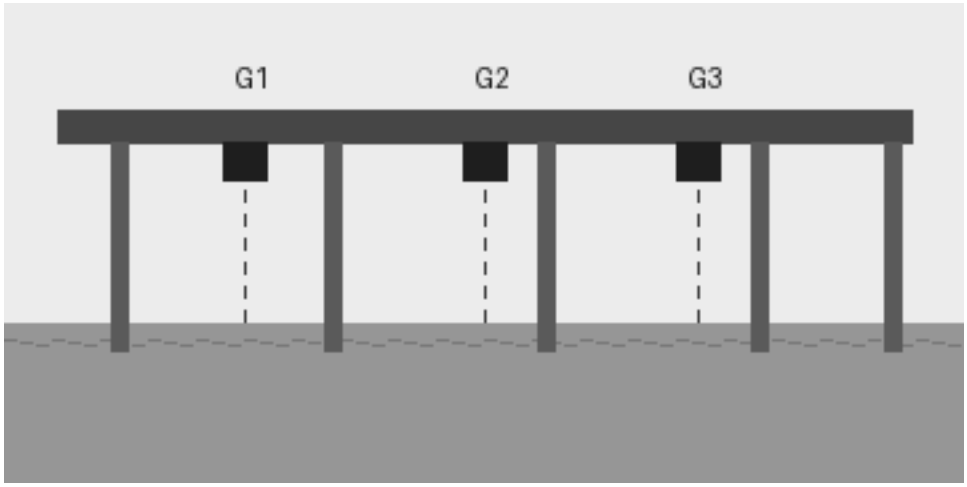

Figure 1. The level gauges on the footbridge, seen from downstream.

3 Findings

- ✔ Gauges 1 and 2 agree with the staff to within 4 mm on every visit.
- ✔ Gauge 3 read 23 mm high after the flood of April 16 and was re-zeroed on May 14.
- ✔ The rating agrees with the gaugings to within 3.1 percent; no change to C is needed.

Field team: J. Körber, A. Nuñez, R. Sándor • resumé of visits on file.

Appendix A Printout of logger 2 on May 14

```
RIVERSIDE WEIR  LOGGER 2  STATION 04117   DUMP 96-05-14 09:12

 TIME   G1MM  G2MM  G3MM  TAIR  BATT  FLAGS
 08:00   639   635   614  11.2  12.6  ......
 08:15   639   636   614  11.4  12.6  ......
 08:30   640   636   615  11.9  12.6  ......
 08:45   640   636   615  12.3  12.6  ......
 09:00   641   637   615  12.8  12.6  ..Z...
 09:15   641   637   638  13.1  12.6  ...R..

 ZERO OFFSET G3 SET +23  OPERATOR JK  OK
 NEXT CALIBRATION DUE 96-08-14
```